

Course Code: CSE 2033	Course Title: Go Programming Type of Course: Theory Only Course	L- T-P- C	3 -0	0	3
Version No.	1.0				
Course Pre-requisites	Computer Programming/ Object Oriented Programming (java)				
Anti-requisites	NIL				
Course Description	Go is an open source programming language created by Google. Go is expressive, concise, clean, and efficient. Its concurrency mechanisms make it easy to write programs that get the most out of multicore and networked machines. Go compiles quickly to machine code yet has the convenience of garbage collection and the power of run-time reflection. It's a fast, statically typed, compiled language that feels like a dynamically typed, interpreted language. It is gaining popularity and it is continuing to grow rapidly in industries such as Dropbox, Uber etc. This course will provide an introduction to the Go programming essentials to students of Engineering through lecture hours with demonstrations. Topics: Topics covered in this course are go program structure; data types and control statements; Composite Types – arrays, slices, strings, runes, bytes, hash maps; functions; methods; garbage collection essentials – pointers, structs, interfaces; error handling; Concurrency – go routines and channels, Packages – import and create custom packages and applications of Go				
Course Objective	The objective of the course is to familiarize the learners with the concepts of GO Programming and attain Employability Skills through Problem Solving techniques.				
Course Out Comes	On successful completion of the course the students shall be able to: CO1: Identify primitive programming constructs in GO. (Knowledge) CO2: Discuss composite data types with concepts of modular programming. (Comprehension) CO3: Implement garbage collection using pointers, structs, interfaces and modules. (Application) CO4: Apply concurrent programming and test routines with applications. (Application)				
Course Content:					
Module 1	Introduction to Go Programming Language	Assignment	Data Collection/Interpretation	10 Sessions	
Topics: [Knowledge]					
Feature of Go language, Installing and Configuring the development environment- Go tools and playground. Structure of Go program; Basic types-numbers, boolean, strings, runes. Variables- declaration, zero values, naming, rules, conversions, constants, multiple variables. Introduction to packages, functions					

from other packages, println, reading input, Control Structures - if, switch, for, programming exercises using control statements.

Module 2	Composite types and functions	Assignment	Data Collection/Interpretation	9 Sessions
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Topics: [Comprehension]

Composite types - arrays, slices, slices with overlapping storage, Structs. Functions-declaring, parameters, returning multiple values, variadic functions; Programming exercises

Module 3	Pointers, Structs, Interfaces and modules	Quiz	Case studies / Case let	9 Sessions
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Topics: [Application]

Pointers: *and & operator, types, pointers with functions, garbage collector – history, Methods and Interfaces, Modules, packages – importing and creating custom packages; Programming exercises.

Module 4	Concurrency and Applications	Quiz	Case studies / Case let	7 Sessions
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Topics: [Application]

Concurrency using Go routines, multiple go routines, channels – channel operations, Testing- writing test, Go test command, Core Packages for – strings, containers and lists, Writing Web Applications, Basic Statistical Computations, histogram plotting, encryption and decryption.

Targeted Application & Tools that can be used:

<https://go.dev/play/>

<https://go.dev/doc/install>

Project work/Assignment:

Text Book

T1 1. John Badner, "Learning Go: An Idiomatic Approach to Real World Go Programming", Oreilly, California, 2021.

References

R1. 1. Alan A.A. Donovan and Brian W. Kernighan, "The Go Programming Language", Pearson Education, India, 2016.

R2. Tsoukalos M. Mastering Go: Create Golang production applications using network libraries, concurrency, machine learning, and advanced data structures. Packt Publishing Ltd; 2019 Aug 29.

Web resources: <https://www.golangprograms.com/go-language.html>

EBSCO database of Presidency University: <https://puniversity.informaticsglobal.com/login>

W3. GO document: <https://go.dev/doc/>

Online tool for program execution:

GO Play Ground - <https://go.dev/play/>

Download and install: <https://go.dev/doc/install>

Topics relevant to development of “Employability”: Go Programming basics for developing Employability Skills through Problem Solving methodologies. This is attained through assessment component mentioned in course handout.